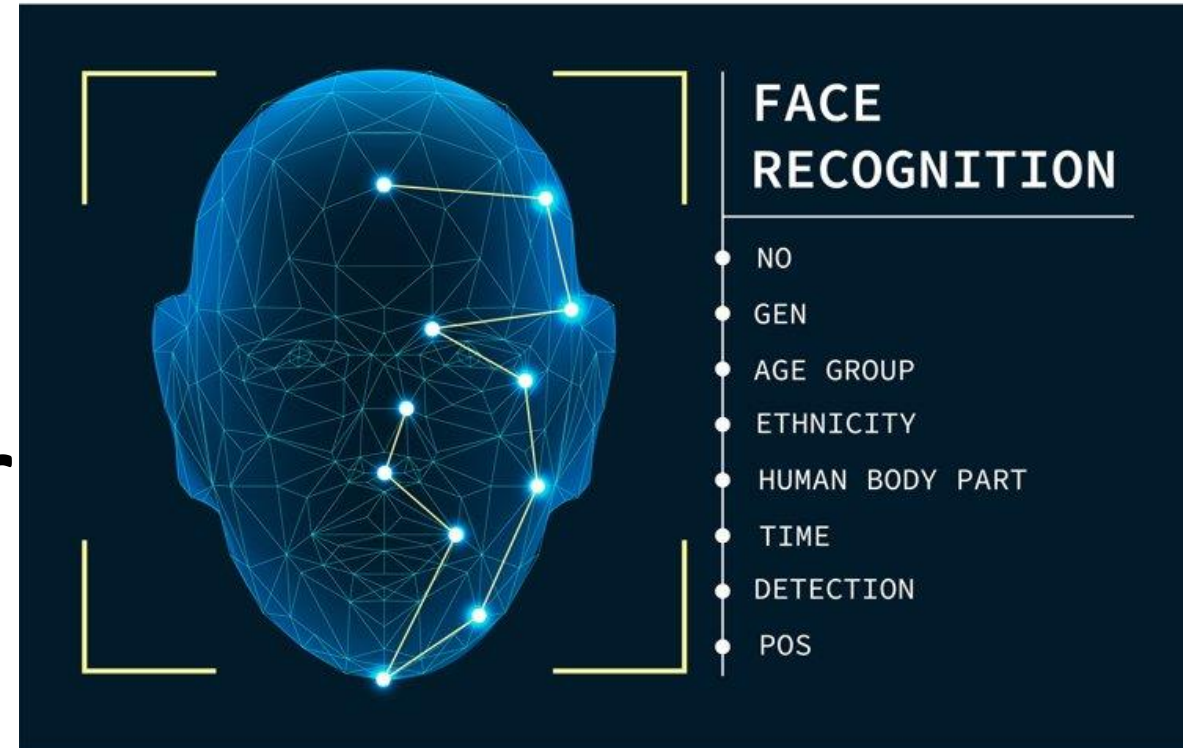


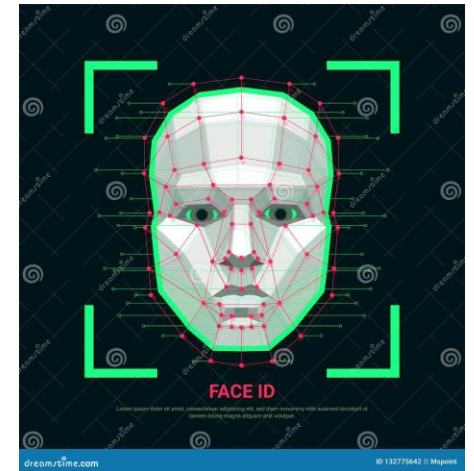
Analyzing Face Recognition System for IoT with Further Improvement

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Introduction

- This presentation focuses on our results of the Face Recognition System.
- Face Recognition is the process of identifying or verifying a person by using their facial features.
- Modern systems rely on extracting unique features and comparing them with stored templates.
- Machine Learning and Artificial Intelligence mainly play huge roles in Face Recognition.
- Our main goal is to check if our designed face recognition system finds low scores for same individual subjects and high scores for different individual subjects after the comparisons of gallery and probe face images of this system.



Our Choice's Programming Language



This project uses Python that is mainly used in Computer Vision.



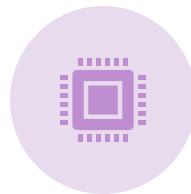
It offers various libraries and functions that allow us to compare gallery and probe face images and generate a score matrix easily.



Hamming Distance and Decidability Index Value are determined by NumPy to find the scores and the index value.



The scores are stored in a score matrix that is organized by Pandas.



Therefore, Python has played a relevant role in making a Face Recognition System.





System_A

- System_A is a simple face recognition system that is based on binary features.
- It uses binarization by thresholding to create a binary image from a grayscale image. Binarize images by using a fixed threshold of 128. A pixel is set to 0 if intensity ≤ 128 or 1 if intensity > 128 .
- Read the grayscale versions of gallery and probe face images that are converted to binary images.
- Comparison between binary images is done by using the Hamming Distance metric before the score are calculated for the score matrix.
- The decidability index value is determined by its formula to see if the system's overall performance is good for the distribution of the scores.



System_A's Results

- The decidability index value of System_A is 2.91 due to the formula of the decidability index value.
- We have compared this index value to another index value of face recognition system FaceRec, and it has 3.5 as the index value.
- The comparison proves that $2.91 < 3.5$, so System_A's performance is not better than FaceRec's performance.
- Because System_A uses binarization that ignores important texture, gradient, and illumination information when the binary images only have black and white colors.
- FaceRec has a high index that allows the system to perform efficiently and find valid subjects. Therefore, System_A is not better than FaceRec.

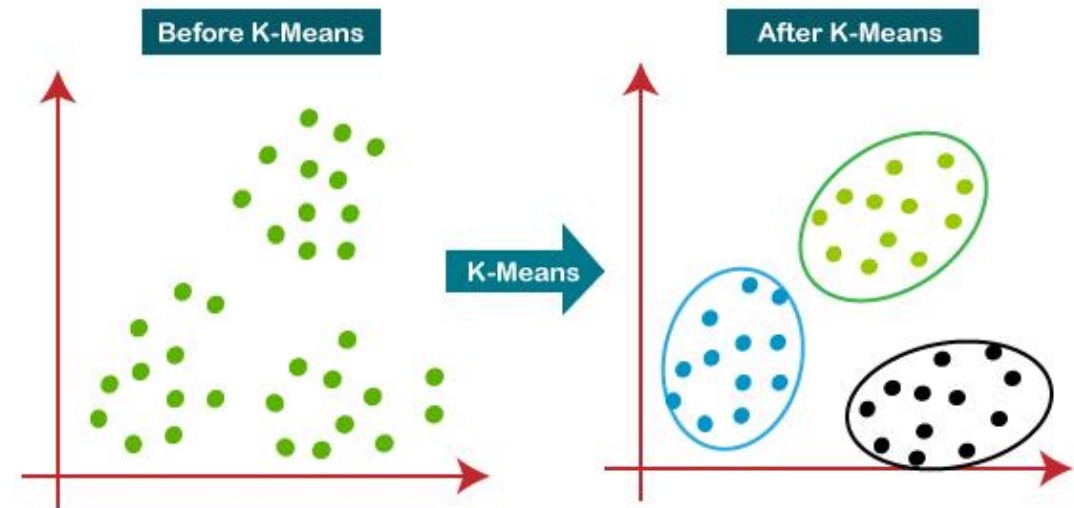


System_B

- System_A is previously used as a simple face recognition system with binary features.
- System_B uses another feature binarization technique to improve the performance of this system.
- But it uses binary features too since System_B will try to enhance the performance of the face recognition system.
- The same rules apply for this system because our goal is to make sure that the output images are the binary images.
- We have decided to use a feature binarization technique of our choice for System_B.

System_B's Proposed Method

- Our proposed technique is the K-Means clustering algorithm that is utilized in System_B to ensure accuracy of gallery and probe face images.
- K-Means algorithm is an unsupervised ML method that partitions unlabeled data into k distinct clusters. We have made 2 clusters of grayscale pixels due to binarization.
- This method eliminates the usage of a threshold that is previously used in System_A for the binarization.
- The cluster with high pixels will change high pixel to 1, and the cluster with low pixels will change low pixel to 0 due to the binarization technique.
- Then, the clustering will convert the grayscale images to the binary images simply since the clusters assign binary pixels to the images.

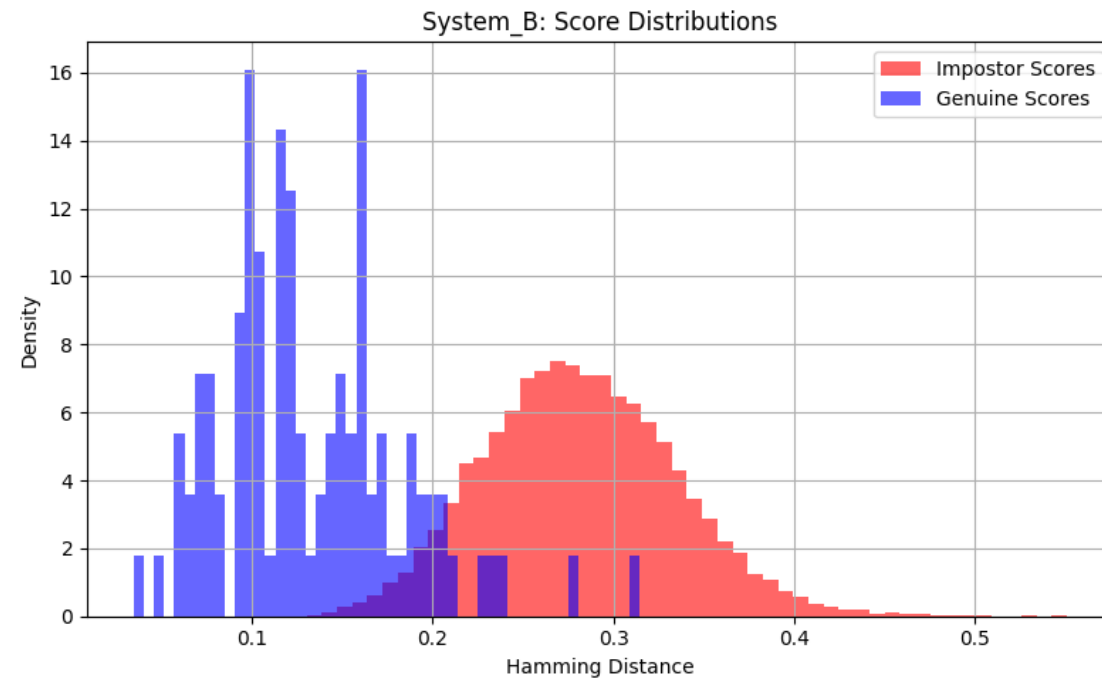




System_B's Results

- The clustering of the pixels has enabled us to generate a score matrix and the decidability index value successfully.
- The decidability index value of System_B is 2.911 due to the formula of the decidability index value.
- K-Means allows us to improve the reliability of the Hamming Distance between gallery and probe set images.
- We have compared the index values of System_A and System_B, and they are explained in the upcoming slide.
- Also, we have generated a histogram that is shown and explained in the next slide.

System_B's Histogram





System_A and System_B's Comparison

- System_A's decidability index value is 2.91, and System_B's decidability index value is 2.911 due to the formula of the decidability index value.
- Then, we have compared these index values by the Improvement Factor or IF.
- We get to find that the value of IF is 0, which proves that there is no improvement of System_B even though we have used K-Means clustering algorithm.
- It proves that System_A's performance is equal to System_B's performance since no improvement is found.
- But there is not negative IF value since it will show that the performance of System_B is worse than the performance of System_A.



Proposed Method's Interpretation

- Interpretation of Results:
- The application of the **K-Means clustering algorithm** was intended to enhance the performance by providing adaptive thresholding for image binarization compared to the fixed thresholding approach used in System_A.
- Despite this methodological improvement, the obtained **decidability index value (2.911)** shows minimal improvement compared to **System_A's index value (2.91)**, indicating virtually equivalent performance.
- Analysis of Minimal Improvement:
- The negligible increase in the decidability index suggests that binarization alone, even with adaptive clustering, remains limited for capturing detailed facial features crucial for robust recognition.
- This underscores the inherent limitations of binary feature representations, which omit nuanced details like textures, gradients, and illumination variations, vital for discriminative performance in facial recognition systems.
- Further Implications:
- Future enhancements might consider incorporating multi-level feature extraction techniques or leveraging advanced deep learning frameworks, which are better suited to capturing complex feature interactions necessary for significant improvements in facial recognition accuracy.



Conclusion

- **Summary of Findings:**
 - Both systems, **System_A** and **System_B**, have been rigorously assessed for their effectiveness in face recognition using binary feature extraction methods.
 - The comparative evaluation, measured by the **decidability index**, indicates no significant performance improvement despite the adoption of a more sophisticated binarization technique (**K-Means clustering**) in System_B.
- **Insights and Limitations:**
 - The results clearly demonstrate that simple binary approaches (fixed thresholding and clustering-based binarization) have inherent limitations, lacking sensitivity to detailed facial attributes essential for discriminative recognition.
 - The negligible improvement factor (**IF = 0**) highlights the constraint of binary representation in achieving high accuracy in face recognition applications, particularly within IoT contexts where environmental conditions vary significantly.
- **Recommendations for Future Work:**
 - Future studies should explore more sophisticated methodologies, such as deep convolutional neural networks (CNNs) or hybrid models combining texture and gradient-based features, to substantially enhance performance.
 - Expanding datasets and incorporating varying lighting conditions, angles, and expressions will further improve robustness and applicability in real-world IoT scenarios.



Thanks for watching our
presentation. Let us know if you
have any questions or doubts
after our presentation.